MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examination
Higher 2

H2 Mathematics 9758/01

Paper 1 **14 September 2018**

3 Hours

Additional Materials: Writing paper
Graph Paper
List of Formulae (MF 26)

READ THESE INSTRUCTIONS FIRST

Write your name and civics group on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
You are expected to use a graphing calculator.
Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.
Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.
You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

1 Express $\dfrac{-x^2+11x-11}{x^2-4x+4}+3$ as a single simplified fraction.

Hence, without using a calculator, solve the inequality

$$\frac{x^2-11x+11}{x^2-4x+4}<3. \qquad [5]$$

2 **(a)** Interpret geometrically what $\mathbf{a}\times\mathbf{b}$ means, given that $\mathbf{a}$ and $\mathbf{b}$ are non-zero and non-parallel vectors. [1]

(b) Show that a formula for the area of triangle OAB can be given as $k\left|\overrightarrow{OA}\times\overrightarrow{OB}\right|$, where k is a constant to be determined. [2]

Hence give the geometrical meaning of $\left|\overrightarrow{OA}\times\overrightarrow{OB}\right|$ in relation to an appropriate quadrilateral. [1]

3 **(i)** Show that $1-\mathrm{e}^{in\theta}=\mathrm{e}^{\frac{1}{2}in\theta}\left(-2\mathrm{i}\sin\dfrac{n\theta}{2}\right)$, and $1+\mathrm{e}^{in\theta}=\mathrm{e}^{\frac{1}{2}in\theta}\left(2\cos\dfrac{n\theta}{2}\right)$ where $n\in$ ¡ .

[2]

(ii) It is given that $z=\mathrm{e}^{\mathrm{i}\theta}$, where $0\le\theta\le\dfrac{\pi}{2}$, using **(i)**, show that

$$\left|1-z+z^2-z^3\right|=4\sin\frac{\theta}{2}\cos\theta. \qquad [4]$$

4 Find

(a) $\int x\sqrt{5-x^2}\ \mathrm{d}x$, [2]

(b) $\int \sin(\ln x)\ \mathrm{d}x$ where $x>0$, [3]

(c) the exact value of $\int_{-\frac{\pi}{4}}^{\frac{\pi}{12}}\cos x\left|\sin x\right|\ \mathrm{d}x$. [4]

5 It is given that

$$\mathrm{f}: x \mapsto \left|\frac{1}{x-3}\right|, \text{ where } x \in \mathbb{R}, x \neq 3,$$
$$\mathrm{g}: x \mapsto \ln x, \text{ where } x \in \mathbb{R}, x > 0.$$

(i) Explain why the composite function gf exists and find gf in a similar form. [3]

(ii) Explain why f does not have an inverse. [1]

(iii) If the domain of f is further restricted to $x < k$, state the maximum value of k such that f^{-1} will exist. Hence find f^{-1} in a similar form. [4]

(iv) State the geometrical relationship between f and f^{-1}. [1]

6 **(a)** **(i)** Using standard series from the List of Formulae (MF26), find the first three non-zero terms of the Maclaurin's series for $y = \dfrac{1}{\sqrt{1+\ln(1+3x)}}$. [3]

(ii) Deduce the approximate value of $\int_0^1 y \, \mathrm{d}x$. Explain why the approximation is not good. [2]

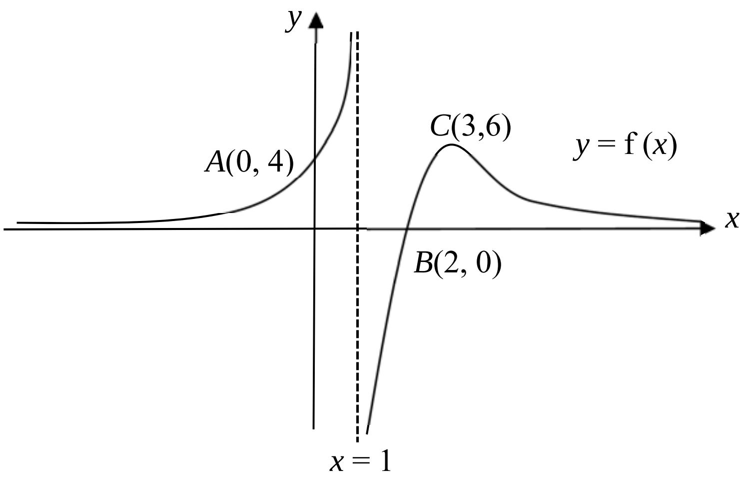
(iii) State the equation of the tangent to the curve $y = \dfrac{1}{\sqrt{1+\ln(1+3x)}}$ at $x = 0$. [1]

(b) Show that, when x is sufficiently small for x^3 and higher powers of x to be neglected,

$$\frac{\cos 2x}{1-\sin x} \approx a + bx + cx^2,$$

where a, b and c are constants to be determined. [3]

7 **(a)**



The diagram above shows the curve $y = \mathrm{f}(x)$ with a maximum point at $C(3, 6)$. The curve crosses the axes at the points $A(0, 4)$ and $B(2, 0)$. The lines $x = 1$ and $y = 0$ are the asymptotes of the curve.

Sketch on separate diagrams, the graphs of

(i) $y = \mathrm{f}'(x)$, [3]

(ii) $y = \dfrac{1}{\mathrm{f}(x)}$, [3]

stating clearly, where applicable, the equations of the asymptotes, the axial intercepts and the coordinates of the points corresponding to A, B and C.

(b) Show that the equation $y = \dfrac{3x^2 - 6x}{x-1}$ can be written as $y = A(x-1) + \dfrac{B}{(x-1)}$, where A and B are constants to be found. Hence state a sequence of transformations that will transform the graph of $y = \dfrac{1}{x} - x$ to the graph of $y = \dfrac{3x^2 - 6x}{x-1}$. [4]

8 The points A and B have position vectors $-2\mathbf{i}-7\mathbf{j}+3\mathbf{k}$ and $\mathbf{j}+\mathbf{k}$ respectively. The plane p has equation $x-y=5$.

(i) Find a vector equation of line l passing through the points A and B. [2]

(ii) Find the acute angle between l and p. [2]

(iii) Verify that the point A lies on the plane p. Given that the point C is the reflection of the point B in the plane p, describe the shape formed by the points A, B and C. [2]

(iv) Find a vector equation of the line which is a reflection of the line l in the plane p. [4]

9 An analyst is studying how the population of bluegill fish in a lake changes over time. By considering their natural birth and death rates, he found that the rate of change of the bluegill fish population is proportional to $6x-x^2$, where x is the population, in thousands, of bluegill fish in the lake at time t years. It is given that the initial population of the bluegill fish in the lake is 12000 and there were 8600 bluegill fish after 1 month.

(i) Show that $x=\dfrac{12\mathrm{e}^{12t\ln\frac{43}{26}}}{2\mathrm{e}^{12t\ln\frac{43}{26}}-1}$. [6]

(ii) Find the time taken for the bluegill fish population to decrease to 75% of its initial population. Leave your answer in years to 3 significant figures. [2]

(iii) Deduce the long term implication on the population of bluegill fish in a lake following this model and state an assumption for this model to hold in the long term. [3]

10 **(a)** The sum of the first n terms of a sequence $\{u_n\}$ is given by $S_n = kn^2 - 3n$, where k is a non-zero real constant.

(i) Prove that the sequence $\{u_n\}$ is an arithmetic sequence. [3]

(ii) Given that u_2, u_3 and u_6 are consecutive terms in a geometric sequence, find the value of k. [3]

(b) A zoology student observes jaguars preying on white-tailed deer in the wild. He observes that when a jaguar spots its prey from a distance of d m away, it starts its chase. At the same time, the white-tailed deer senses danger and starts escaping.

He models the predator-prey movements as follows:

The jaguar starts its chase with a leap distance of 6 m. Subsequently, each leap covers a distance of 0.1 m less than its preceding leap.

The white-tailed deer starts its escape with a leap distance of 9 m. Subsequently, each leap covers a distance of 5% less than its preceding leap.

(i) Find the total distance travelled by a white-tailed deer after n leaps. Deduce the maximum distance travelled by a white-tailed deer. [3]

(ii) Assume that both predator and prey complete the same number of leaps in the same duration of time. Given that $d = 11$ m, find the least value of n for a jaguar to catch a white-tailed deer within n leaps. [3]

11 A curve C has parametric equations

$$x = 1 - \cos^3\theta, \qquad y = 1 - 3\sin\theta\cos^2\theta, \qquad \text{for } 0 \le \theta \le \frac{1}{2}\pi.$$

(i) Show that $\dfrac{dy}{dx} = 2\tan\theta - \cot\theta$. [3]

(ii) Show that the exact coordinates of the turning point is at $\left(1-\left(\sqrt{\dfrac{2}{3}}\right)^3,\ 1-\dfrac{2}{\sqrt{3}}\right)$ and explain why it is a minimum. [6]

(iii) Show that the equation of the normal to the curve at $\theta = \dfrac{\pi}{6}$ is $y = 1.73205x - 0.73205$ (rounded off to 5 decimal places) and hence evaluate the area of the region bounded by C, the y-axis and the normal to the curve at $\theta = \dfrac{\pi}{6}$. [6]

End of Paper